

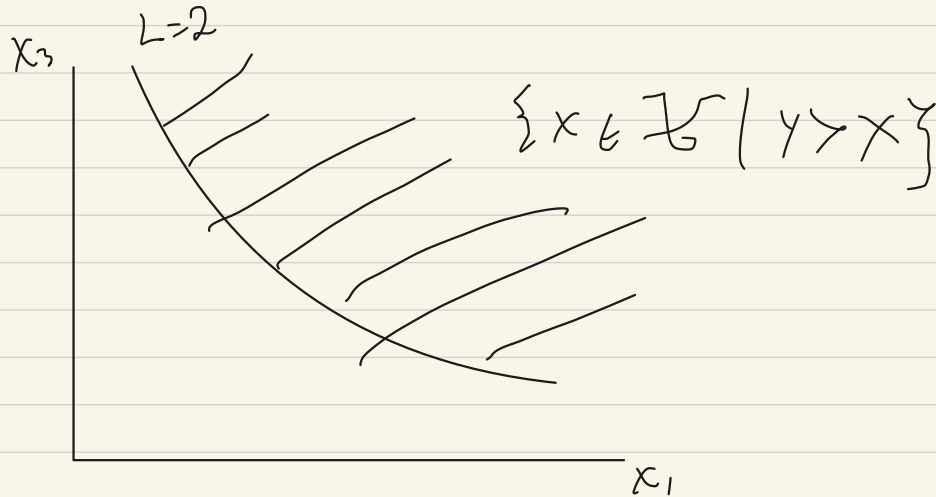
Lesson 7



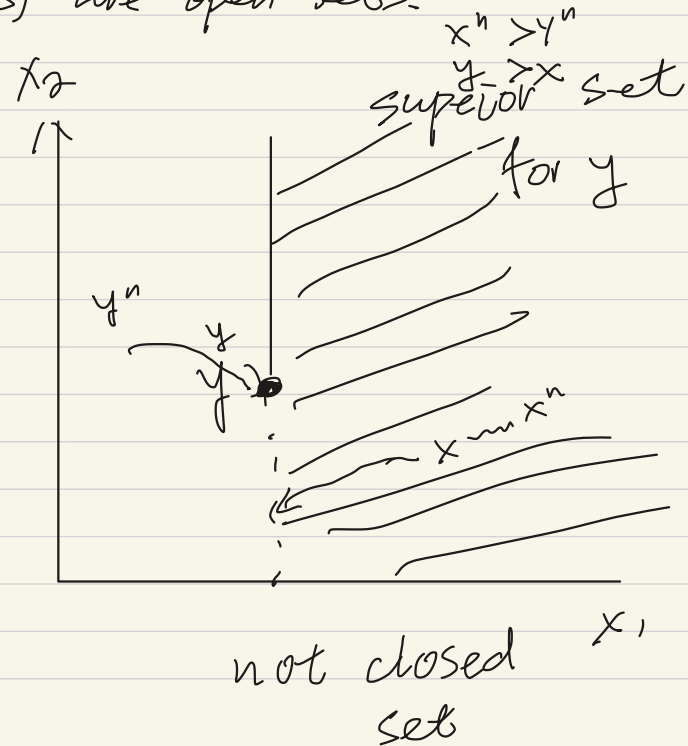
Continuity has the implication that no preference reversal

An equivalent definition: The superior and inferior sets are closed. The complement

(the set with strict preference are open sets) are open sets.



no dramatic change in order
if y changes a bit



The main conclusion is that $\exists$ a continuous utility function that represents $\succeq$ if $\succeq$ continuous

There's no utility function represent lexico-pref.

We need extra assumption that $V(x)$ is differentiable
rule out LEONTIF pref. $V(x_1, x_2) = \min\{x_1, x_2\}$

Other connections between $\succeq$ and $V(x)$
 $\succeq$ is convex iff $V(x)$ is quasiconcave

Utility maximization

Assume $V(x)$ exists and is continuous. Assume preferences are ^{locally} non-satiated!
the utility max prob.

$$\max_{x \geq 0} V(x)$$

$$\text{s.t. } Px \leq w$$

A solution exists since $V(x)$ is continuous and budget set

is non-empty and compact

The solution is Marshallian or Walrasian demand

We may have many solutions in which we have demand
correspondence

Some properties:

- Walra's law: The budget bind due to local-non satiation
- Homogeneity of degree zero $x(p, w)$. The feasible set doesn't change.

The indirect utility function is the max value function for the prob.

$$V(p, w) = V(x(p, w))$$

the highest utility one can achieve.

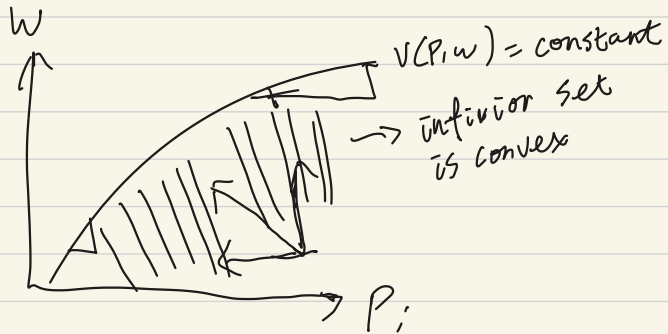
- $V(P, w)$ is continuous (P, w) as a consequence of the more general theorem of the maximum.

- $V(P, w)$ is homogeneity degree zero (P, w)

because the feasible set didn't change.

- $V(P, w)$ is strictly $\uparrow$ in w and non-increasing in $P_i \forall i = 1, \dots, L$.

- $V(P, w)$ is quasi-convex.



I need to be compensated more
for a price change the lower
price was to stand with

I'm pose more structure by assuming $\frac{V(x)}{\text{exogenous}}$ is strictly
quasi concave. $\Rightarrow$ unique solution, $x(p, w)$ is a demand function

Let us also assume $x(p, w)$ and $V(p, w)$
endogenous

are differentiable A crm.

The good news we can now use the wonderful
And elegant theorem envelope theorem.

E.V. Theorem: what happen to max val. indirect utility. When a parameter change.

$$\frac{\partial V(p, w)}{\partial w} = \frac{\partial \lambda}{\partial w} \bigg|_{\substack{x=x(p, w) \\ \lambda=\lambda^*}} = \lambda^* > 0$$

$$d = V(x) - \lambda(p \cdot x - w) \quad \checkmark \quad \begin{matrix} \max V(x) \\ p \cdot x \leq w \end{matrix}$$

$$\frac{\partial v(p, w)}{\partial p_i} = \frac{\partial \mathcal{L}}{\partial p_i} \Big|_{\substack{x=x(p, w) \\ \lambda=\lambda^*}} = -\lambda^* x_i(p, w) \leq 0$$

combining two gives Roy's identity

$$x_i(p, w) = \frac{-\frac{\partial v(p, w)}{\partial p_i}}{\frac{\partial v(p, w)}{\partial w}}, \text{ so we can infer demand from the indirect utility function.}$$

Starting from the Lagrangian and assume the solution is interior $x \gg 0$, then the KKT conditions are

$$\frac{\partial v(x)}{\partial x_i} - \lambda p_i = 0 \Leftrightarrow \frac{\partial v(x)}{\partial x_i} = \lambda p_i$$

if $\frac{\partial V(x_i)}{\partial x_i} > 0$ or $\nabla V(x) > 0$ then $\lambda^* > 0$. Constraint is binding!

Divide for x_i and x_j

$$\frac{\frac{\partial V(x)}{\partial x_i}}{\frac{\partial V(x)}{\partial x_j}} = \frac{p_i}{p_j}$$

marginal rate of substitution = relative price

this is the tangent condition.

Expenditure mini:

A thought experiment

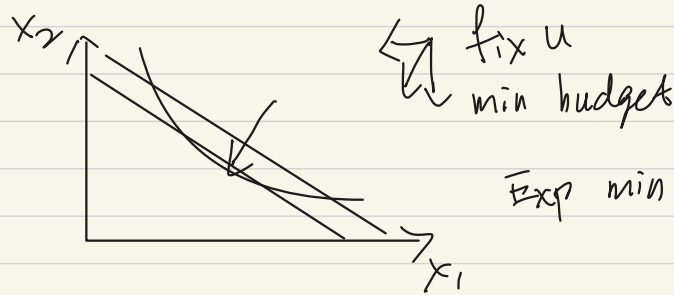
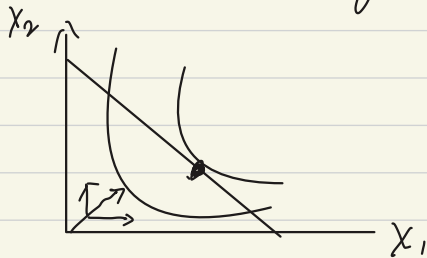
$$\min P \cdot x$$

$$x \geq 0$$

$$\text{s.t. } V(x) \geq U$$

heapest way of getting utility no less than U

The dual of utility max. we switched objective and constraint.



The expenditure func. is the minimum value func.

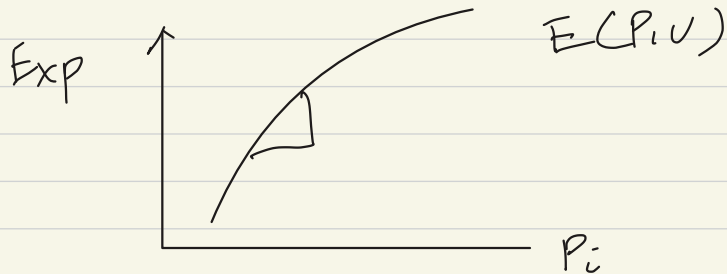
$$E(P, u)$$

- properties
- $E(P, u)$ condition on (P, u)
 - $E(P, u)$ is homogeneous degree 1.

relative prices are unchanged, solution is unchanged

But the bundle now costs more. $E(\alpha P, u) = \alpha E(P, u)$, $\alpha > 0$

- $E(P, u)$ is non-decreasing in u and in P_i , $i = 1, \dots, L$.
- $E(P, u)$ is concave on P .



P.f. Let $P'' = \alpha P + (1-\alpha)P'$, $\alpha \in [0, 1]$

Let x'' denote the solution for expend.

min. prob. Then

$$\begin{aligned} E(P'', v) &= P'' \cdot x'' = \alpha P \cdot x'' + (1-\alpha)P' \cdot x'' \\ &\geq \alpha E(P, v) + (1-\alpha) E(P', v) \end{aligned}$$

confirming concavity.